

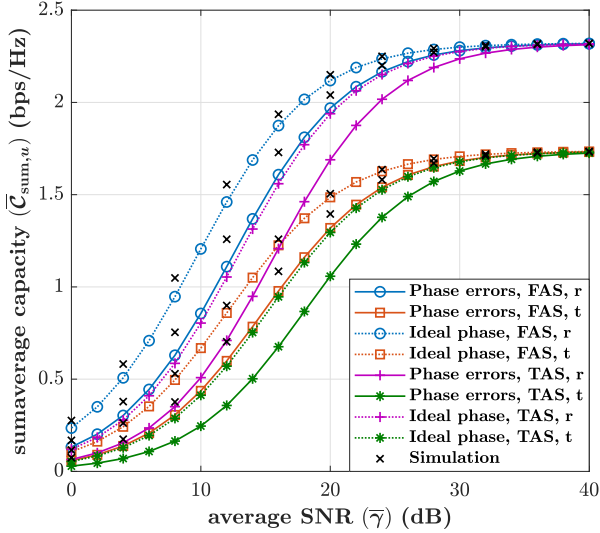


Fig. 8. The sum AC results of FAS-assisted STAR-RIS RSMA versus the average SNR $\bar{\gamma}$ with phase errors and ideal phases for $K = 30$.

especially in channels with rich scattering.

The results in Fig. 8 study the impact of phase errors on the sum AC, $\bar{C}_{\text{sum},u} = \bar{C}_{c,u} + \bar{C}_{p,u}$. First of all, as expected, a small gap between the simulation results and the theoretical predictions is observed. This discrepancy arises from the use of a heuristic technique to approximate the expectation of the maximum of N_u correlated Gamma RVs, as discussed in Remark 2. The heuristic method, based on simplifying assumptions, provides a computationally efficient approximation that captures the essential behavior of the system, especially in cases involving moderate correlations. Despite the gap, the results indicate that this approach performs well across a range of scenarios, making it a useful approximation for evaluating the AC performance under correlated conditions. For both FAS and TAS, we see that as $\bar{\gamma}$ increases, the sum AC initially improves but eventually saturates. This occurs because in the RSMA scheme, rate splitting effectively manages interference and optimizes power allocation at moderate SNR levels. As $\bar{\gamma}$ grows, the SINR expressions in (8) and (11) converge to deterministic values, depending on the power allocation factors for the common and private streams. Beyond this threshold, further increases in SNR do not significantly improve the sum AC, resulting in saturation. Additionally, the case with ideal phase results in a higher level of sum AC compared to the scenario with phase errors. This is because with ideal phase, STAR-RIS can perfectly align the transmitted signals with the desired reflection coefficients, thereby maximizing the signal power at the receiver and minimizing interference. In contrast, phase errors introduce misalignment between the transmitter and STAR-RIS, leading to suboptimal signal reflection and increased interference. As a result, the effective channel gains are reduced in the presence of phase errors, causing a degradation in the sum AC. Noteworthy, the asymptotic behavior for the FAS and TAS schemes is independent of the phase errors, so does the performance degradation because the phase

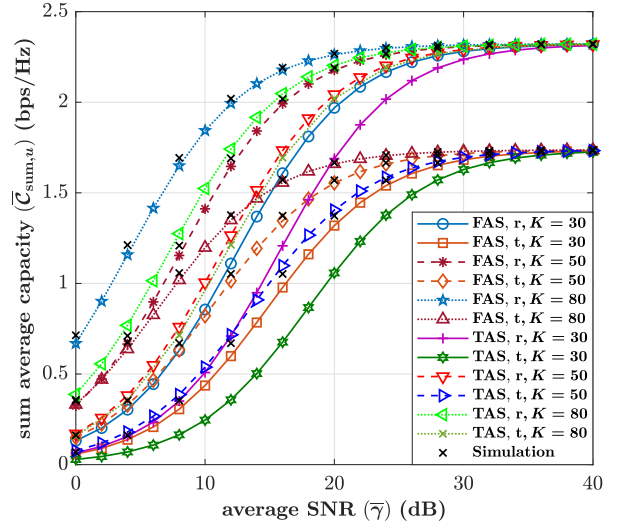


Fig. 9. The sum AC results of FAS-assisted STAR-RIS RSMA versus the average SNR $\bar{\gamma}$ with phase errors for different numbers of STAR-RIS elements K .

mismatch is reduced as $\bar{\gamma}$ grows.

Finally, Fig. 9 studies how the number of STAR-RIS elements, K , affects the sum AC. As we can see, a larger K provides a higher sum AC for both FAS and TAS. Increasing K also results in a higher sum AC for both FAS and TAS. This gain occurs because a larger number of STAR-RIS elements increases the surface's ability to adaptively reconfigure the channel, allowing for better alignment of the reflected signals. With more elements, STAR-RIS can have more precise control over the spatial distribution of the reflected signals, effectively enhancing the SNR and reducing interference. Furthermore, a larger K enables better exploitation of spatial diversity, thereby improving the capacity for both RSMA users. As such, the increase in the number of STAR-RIS elements contributes to a more efficient channel utilization, leading to a higher sum AC. We can also see that FAS provides a higher sum AC than the TAS counterpart. This is because FAS can dynamically adjust its configuration, improving channel alignment and spatial diversity, while TAS, with its fixed setup, lacks this flexibility, limiting its performance in varying conditions. As in the previous figure, the sum AC saturates to a value that only depends on the RSMA power allocation factors, as the dependence on K is cancelled out for sufficiently large $\bar{\gamma}$.

V. CONCLUSION

This paper presented a comprehensive investigation of FAS-assisted STAR-RIS communication systems employing RSMA signaling, with a particular focus on the impact of phase errors under the ES protocol. By modeling the phase errors with a generic distribution and deriving the equivalent channel gain characterized by the multivariate t -distribution, we provided a realistic and robust analysis of the system performance. Compact analytical expressions for key performance metrics, such as outage OP and AC were derived, with the latter being